

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AB2-074***

Sunbury #2 500kV

May 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Shamokin Dam Station LLC, the Interconnection Customer (IC), has proposed an upgrade to their planned natural gas fueled generating facility to be located in Snyder County, Pennsylvania. The proposed upgrade rate is for 113 MWs Energy (MWE) and 50 MWs Capacity (MWC). Note that this project is an increase to the IC's AA2-182 project, which will share the same site and Point of Interconnection (POI). The AA2-182 project has a capability of 1030 MWE and 977 MWC. The total capability of the combined AA2-182 and AB2-074 projects will be 1143 MWE and 1027 MWC. The proposed in-service date for the AB2-074 project is September 2019. **This study does not imply a PPL Electric Utilities Corporation (PPL EU) commitment to this in-service date.**

Point of Interconnection

AB2-074 will interconnect with the PPL EU transmission system at the same POI as the AA2-182 project at the Sunbury 500kV substation (See Attachment 1).

Cost Summary

The AB2-074 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 0

Description	Total Cost
Total Costs	\$ 0

In addition, the AB2-074 project may be responsible for a contribution to the following costs:

Description	Total Cost
New System Upgrades	\$ 0
Previously Identified Upgrades	\$ 0
Total Costs	\$ 0

Attachment Facilities

There are no additional Attachment Facilities required by PPL EU to support the increase in output from this facility.

Direct Connection Cost Estimate

There are no additional Direct Connection Facilities required by PPL EU to support the increase in output from this facility.

Non-Direct Connection Cost Estimate

There are no additional Non-Direct Connection Facilities required by PPL EU to support the increase in output from this facility.

Interconnection Customer Requirements

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

Network Impacts

The Queue Project AB2-074 was evaluated as a 170.0 MW (Capacity 170.0 MW) uprate to the AA2-182 project at the Sunbury 500kV substation in the PPL area. Project AB2-074 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AB2-074 was studied with a commercial probability of 53%. Potential network impacts were as follows:

Base Case Used

Summer Peak Analysis – 2020 Case

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Short Circuit

No issues

Stability and Reactive Power Requirement

The Stability Analysis is currently in progress under the AA2-182 queue project.

Affected System Analysis & Mitigation

No affected systems identified.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis – 2020 Case

None

New System Reinforcements

None

Contribution to Previously Identified System Reinforcements

None

Potential Issues

PPL indicates there are known stability concerns in this area.

Attachment 1

Sunbury 500 kV GIS AA2-182 & AB2-074

